

How is the ESC for motor constructed? Is there any such article published in EFY using Arduino or STM8s or STM32F?

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ESC, or electronic speed control, is a circuit used to change the speed of an electric motor. There are two types of ESCs—brushed and brushless. Brushed ESC is used for a brush motor, brushless ESC is used for a brushless motor. Depending on the type, their construction and design are different. Brushless ESCs are more advanced and costly than the brushed ESCs.

The construction of most ESC modules includes a power supply stage, a current-sensing circuit, a controller, and a communication interface. In order to design an ESC, you need to know the position of the rotor. There are two common methods used for determining the rotor position—Hall-effect sensors embedded in stator and sensing the back electromotive force from motor. The ESC is used to capture these signals.

The design and construction of brushless ESC can be done using 3-phase bridge drivers such as IRS2330, L6234, or microcontrollers. Designing an ESC for drones requires high-quality components that are designed to control high-RPM motors. There are some microcontrollers specifically designed for such applications. For example, Texas Instruments has a family of MCUs, called InstaSPIN, that simplify the design of three-phase motor control applications. STMicroelectronics offers a complete ESC reference design, implementing a sensorless FOC (field-oriented control) algorithm.

Most of the modern electronic speed controllers include a microcontroller to process the input signal and control the electric motor through an inbuilt program. A simplified ESC block diagram is shown in Fig. 1.

The construction of ESC has not been covered in EFY. However, the design and constructions of ESCs using

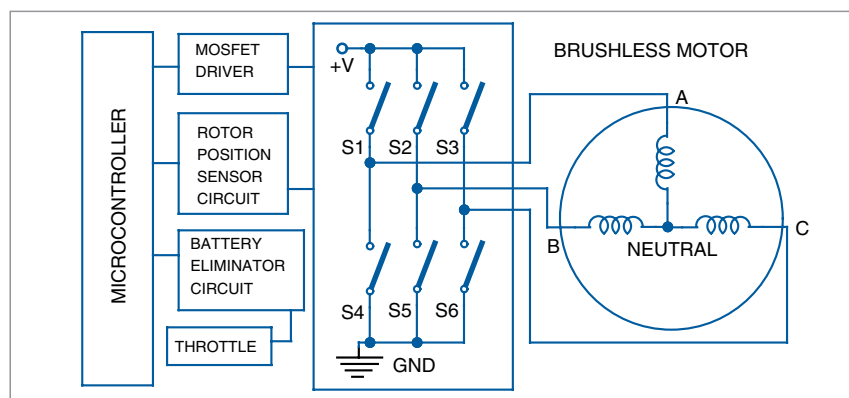


Fig. 1: Simplified ESC block diagram

various techniques including Arduino can be found at following websites:

<https://www.powerelectronicsnews.com/designing-an-esc-module-to-control-drone-motors>

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How to select fuse across the primary or secondary side of a transformer and how to calculate ampere rating of the fuse?

Anirvan

There are different types of fuses having different characteristics. Most common types are categorised as fast-acting and slow-acting fuses. Fuse characteristics can be adapted to different loads. Common fuse types include glass-tube fuses, miniature circuit-breakers, thermal fuses.

While selecting a fuse, current rating is more important than voltage rating; a fuse can operate at any voltage up to its rated voltage. Choose a standard fuse size in relation to the current specified on the transformer. The rule is to find the nearest standard fuse that is rated larger than the rated current.

The secondary fuse is placed after the transformer's output points of connection on the secondary side. It acts as overload protection for the transformer. Its rating must not exceed the secondary current specified on the transformer. The choice of characteristic, slow or fast, is determined by the load.

The primary fuse is placed in front of the transformer's points of connection on the primary side. It acts as short-circuit protection for the transformer. All distribution transformers have fuses on primary side but low-power transformers for electronic circuitry have optional fuses depending on the applications.

When device is switched on, the starting current can be many times higher than nominal current for a short period. It is important that the primary fuse is as slow-acting as possible.

Suppose a transformer is rated at 230V AC primary and 12V, 1A secondary. The voltage turn ratio is given by primary voltage/secondary voltage = $230\text{V}/12\text{V} = 19.2\text{A}$. The primary fuse current can be calculated from current turn ratio, which is secondary rated current/primary current, that is 0.052A.

You can use a primary fuse that is 135% of this value, which is about 100mA. You can select a nearest standard size fuse with current rating not less than 100mA. Note that you can use this fuse across the secondary side but not after the rectifier circuit. A fuse installed in an AC circuit performs differently when installed in a DC circuit.

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